

NORMAL / VERIFIED

Recommendation: APPROVED: TRANSACTION CLEARED BY RISK PROTOCOLS

RISK SCORE

22.32 / 100

MODEL DECISION

NORMAL

Transaction Overview & Account Baseline

CURRENT TRANSACTION		EVALUATED VALUES	CUSTOMER PROFILE BASELINE	HISTORICAL NORMS
Transaction ID	TXN0000042122		Historical Total Txns	0
Customer ID	CUST00148080		Average Spend	\$143.42
Timestamp	2024-12-30 23:47:00		Spend Std Deviation	\$97.22
Amount	\$103.14 USD		Median Spend	\$103.14
Location	Munich, India		Home City	Munich
Category	Crypto Exchange		Home Country	USA
Payment Method	Debit Card		Primary Category	ATM Withdrawal
Device Type	Mobile		Primary Payment	Credit Card
Distance from Home	13.7 km		Primary Device	Desktop
Time Since Prior Txn	16.01 hrs		Avg Transaction Hour	10.2:00 hrs
Failed PIN Attempts	0		Txn Velocity	0.00 / day
Recent PIN Change	No		Customer Age	18 yrs
Cross-Border / Intl	No (Domestic)		Credit Score	795
Night Window	Yes (Off-hours)		Prior Confirmed Frauds	0

AI Model Risk Breakdown & Explainability

Autoencoder Anomaly Score		0.3513 (0.00 – 1.00)		Reconstruction MSE Error	0.416578 (Baseline Threshold: 0.5929)
RISK COMPONENT	SCORE	WEIGHT	CONTRIB (pts)	AI EXPLANATION & CONTEXT	
Autoencoder	0.35	35%	+12.29	Transaction pattern appears relatively normal (autoencoder score: 0.35).	
Time Anomaly	0.70	10%	+7.00	Transaction at 23:00 — unusual off-hours activity (night transaction flag set).	
Velocity Risk	0.18	10%	+1.83	Transaction 16.0 h after last — within normal velocity range.	
Behavioral Deviation	0.15	8%	+1.20	Behavioural deviations detected: unusual category 'Crypto Exchange', unusual payment method 'Debit Card', unusual device 'Mobile'. Customer normally uses Credit Card in Munich at avg \$143.	
Amount Deviation	0.00	15%	+0.00	Transaction amount \$103.14 is 0.4σ below customer's average \$143.42 (σ=\$97.22).	
Geographic Risk	0.00	15%	+0.00	Transaction in expected city (Munich). No geographic risk.	
International Flag	0.00	3%	+0.00	Domestic transaction — no international risk.	
Failed Attempts	0.00	2%	+0.00	No prior failed attempts on this transaction.	
Pin Change	0.00	1%	+0.00	No recent PIN change.	
Credit Factor	0.00	1%	+0.00	Excellent credit score (795) — minimal credit risk.	

Audit & Compliance Sign-Off**Investigator Signature:****Verified Date:** 2026-10-07

This report is automatically synthesized by the SENTINEL Deep Autoencoder engine with 10-factor weighted anomaly scoring. All data points conform to regulatory audit guidelines.